

Chapter

Human Growth and Development

1. The stage from the age of 12 years to 19 years is called: MAIN 2024

- (a) Childhood Stage
- (b) Infancy Stage
- (c) Adolescence Stage
- (d) Adulthood Stage

2. The process that starts from infancy and continues into late adolescence, concentrating on gross and fine motor skills, is known as: SQP 2021

- (a) Physical Development
- (b) Psychological Development
- (c) Social Development
- (d) Emotional Development

3. At what point in a person's growth and development does he or she become self-directed and experienced? COMP 2023

- (a) Infancy Stage
- (b) Adulthood Stage
- (c) Childhood Stage
- (d) Adolescence Stage

4. Genes and chromosomes affect which factor of human growth and development? SQP 2022

- (a) Heredity
- (b) Environment
- (c) Gender
- (d) Nutrition

5. Why is exercise important for development? SQP 2024

- (a) Only for children
- (b) Helps in physical, emotional, and mental wellbeing
- (c) It only increases muscle strength
- (d) It does not affect development

6. What happens when a child is deprived of proper nutrition? COMP 2021

- (a) Growth is stunted
- (b) Cognitive abilities improve
- (c) Emotional development increases
- (d) No effect on development

7. Assertion (A) : Growth is a finite process that focuses only on physical changes in the body.

Reason (R) : Growth primarily occurs due to cell division and is limited to measurable changes like height and weight. **MAIN 2024**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

8. Assertion (A) : Development continues throughout life and includes physical, emotional, and cognitive changes.

Reason (R) : Development is limited to early stages of life and stops after adolescence. **SQP 2021**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

9. Assertion (A) : Nutrition is a critical factor for healthy growth and development.

Reason (R) : Malnutrition can lead to stunted growth, cognitive issues, and weakened immunity. **COMP 2023**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

10. Assertion (A) : Heredity determines an individual's potential for growth.

Reason (R) : Environmental factors such as stress and pollution do not influence growth and development. **SQP 2022**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

11. Match the following

SQP 2021

Column I (Stages of Growth)		Column II (Characteristics)	
(I)	Infancy	1.	Rapid brain development, motor skills improvement.
(II)	Childhood	2.	Logical thinking, self-control, neuromuscular coordination.
(III)	Adolescence	3.	Puberty, identity exploration, emotional shifts.
(IV)	Adulthood	4.	Stability, maturity, focus on responsibilities.

- (a) I-2, II-1, III-4, IV-3
- (b) I-1, II-2, III-3, IV-4
- (c) I-3, II-4, III-2, IV-1
- (d) I-4, II-3, III-1, IV-2

12. Match the following:

COMP 2023

Column I (Factors)		Column II (Influence)	
(I)	Heredity	1.	Determines genetic potential for growth.
(II)	Gender	2.	Hormonal differences influence growth patterns.
(III)	Nationality	3.	Cultural traditions, lifestyle impact growth.
(IV)	Stress	4.	Can negatively affect physical and cognitive development.

- (a) I-2, II-1, III-4, IV-3
- (b) I-3, II-4, III-2, IV-1
- (c) I-1, II-2, III-3, IV-4
- (d) I-4, II-3, III-1, IV-2

13. Match the following:

SQP 2022

Column I (Concepts)		Column II (Explanation)	
(I)	Quantitative	1.	Measurable changes like height and weight.
(II)	Qualitative	2.	Changes in personality and cognitive skills.
(III)	Growth	3.	Finite process limited to early life stages.
(IV)	Development	4.	Continuous process throughout life.

- (a) I-2, II-1, III-4, IV-3
- (b) I-1, II-2, III-3, IV-4
- (c) I-3, II-4, III-1, IV-2
- (d) I-4, II-3, III-2, IV-1

14. Define Adolescence. MAIN 2024
15. What is infancy? SQP 2021
16. What is the period of childhood? COMP 2023
17. Define Growth. SQP 2022
18. Explain developmental characteristics during childhood. COMP 2024
19. Explain the developmental characteristics during infancy. MAIN 2021
20. "Adolescence is the age of stress and strain." Explain. SQP 2022
21. Rahul, a 12-year-old boy, has difficulty making friends and often feels uncomfortable in social settings. His teacher encourages him to participate in group activities and discussions to help him interact with his classmates.
- (i) Which aspect of development is being focused on?
- (ii) Which skill is being targeted for improvement? MAIN 2024
22. Meera, a 5-year-old child, is learning to hold a pencil and write letters. Initially, she struggles with hand control, but with regular practice, she improves her ability to write neatly.
- (i) Which aspect of development is being enhanced?
- (ii) Which motor skill is being developed? SQP 2021
23. Vikram, a 17-year-old boy, experiences rapid height growth and voice changes. He also starts developing facial hair and becomes more conscious of his appearance.
- (i) Which stage of development is Vikram in?
- (ii) Which type of growth is he experiencing? COMP 2023

Solution

1. (c) Adolescence Stage

Adolescence is the transition period between childhood and adulthood, typically from ages 12 to 19, marked by rapid physical, mental, and emotional changes.

2. (a) Physical Development

Physical development begins in infancy and continues through late adolescence, focusing on the growth of gross and fine motor skills, muscle control, physical coordination, and changes during puberty.

3. (b) Adulthood Stage

In adulthood, a person becomes self-directed and experienced, having reached full physical, mental, and emotional maturity.

4. (a) Heredity

Genes and chromosomes determine inherited traits, so heredity plays a key role in normal physical growth and development during childhood.

5. (b) Helps in physical, emotional, and mental wellbeing

Regular exercise enhances physical fitness, strengthens muscles, improves mental abilities, and supports emotional stability, contributing to overall development.

6. (a) Growth is stunted

Explanation: Malnutrition hampers physical and cognitive growth, leading to stunted development and affecting overall health and learning abilities.

7. (a) Both A and R are true, and R is the correct explanation of A.

Growth involves measurable physical changes such as height, weight, and organ size, which result from cell division, making it a finite process.

8. (c) A is true, but R is false.

Development is a lifelong process that includes quantitative and qualitative changes in physical, emotional, and cognitive aspects. It does not stop after adolescence.

9. (a) Both A and R are true, and R is the correct explanation of A.

Explanation: Proper nutrition is essential for healthy growth and development, while malnutrition negatively affects physical growth, cognitive abilities, and immunity.

10. (c) A is true, but R is false.

11. (b) I-1, II-2, III-3, IV-4

Infancy is marked by brain growth and motor skills. Childhood enhances reasoning and muscle control. Adolescence includes puberty and identity challenges. Adulthood brings stability and responsibility. Each stage represents a different aspect of human growth and development.

12. (c) I-1, II-2, III-3, IV-4

Heredity passes genetic traits from parents to offspring. Gender impacts growth due to hormonal changes. Nationality affects growth through culture, diet, and lifestyle. Stress can negatively influence physical and mental development, affecting overall health and well-being.

13. (b) I-1, II-2, III-3, IV-4

Quantitative changes are measurable, like height and weight, while qualitative changes include personality and intelligence. Growth is a finite process seen in early life, whereas development is an ongoing process throughout a person's lifetime.

14. Adolescence is the transitional period between childhood and adulthood, usually ranging from **12/13 years to 18/19 years**, marked by rapid **physical, emotional, and cognitive growth**. It begins and ends gradually, without fixed dates. This stage is characterized by significant changes in the **endocrine**

(hormonal) system, including the sex glands and other internal secretion glands, leading to puberty-related developments.

15. Infancy is the stage of life from **birth up to 5 years of age**.

16. Childhood is the stage that begins at **5 years of age** and continues until **12 years of age**.

17. Growth is the progressive increase in the **size of a child or its parts**, including **height, weight, body size, and the development of organs** such as the brain.

18. Childhood, from 5 to 10 years, is important for **physical, intellectual, and emotional growth**:

(i) Physical Development:

1. Steady increase in **height and weight**.
2. **Temporary teeth** fall out and are replaced by permanent teeth; girls often grow faster than boys.
3. **Bones and muscles** develop, improving coordination for activities like running and jumping.

(ii) Intellectual Development:

1. **Language skills** improve, enabling better communication.
2. Memory, reasoning, and **observation abilities** develop.
3. Children explore their surroundings, showing **curiosity** and interest in learning.

(iii) Emotional Development:

1. Emotional sensitivity increases; children understand and express feelings like **empathy and guilt**.
2. They become aware of their **actions and consequences**.
3. This stage is key for building **self-esteem** and learning to manage emotions.

19. Infancy, from birth to 5 years, is a period of rapid physical, neurological, and emotional development:

(i) Physical Growth:

1. Significant increase in **height and weight** with rapid **bone and muscle development**.
2. Acquisition of **basic motor skills** such as crawling, sitting, standing, and walking.

(ii) Neurological Development:

1. Early stages are dominated by **reflex actions** like sucking, crying, and eye movements.
2. Gradual improvement in **coordination** between sensory organs and motor functions, allowing grasping, rolling, and walking.

(iii) Social and Emotional Growth:

1. Infants form **bonds with caregivers** and express emotions like joy, fear, and anger.
2. Development of **trust and security** occurs through consistent care and attention.

20. Adolescence, called the “age of stress and strain,” is a **transitional phase from childhood to adulthood** marked by rapid physical, emotional, and social changes:

(i) Physical Stress:

1. Rapid **growth spurts**, weight changes, and development of secondary sexual characteristics.
2. These changes can cause **awkwardness** and self-consciousness about appearance.

(ii) Emotional Strain:

1. Hormonal changes lead to **mood swings**, irritability, and heightened emotional sensitivity.
2. Adolescents may feel **anxious** about their identity, independence, and future.

(iii) Social Pressure:

1. **Peer pressure** and societal expectations increase.
2. Balancing personal desires with societal norms can cause **conflicts and stress**.

(iv) Psychological Adjustments:

1. Challenges in developing **decision-making skills** and handling responsibilities.
2. Poor coping mechanisms may result in **risky behavior** or social withdrawal.

(v) Coping and Guidance:

1. Proper support from parents, teachers, and peers is essential.
2. **Open communication, emotional support, and a positive environment** help reduce stress and strain during adolescence.

21. (i) Social Development

(ii) Interpersonal and Communication Skills

22. (i) Physical Development

(ii) Fine Motor Skills

23. (i) Adolescence (ii) Physical Growth (Puberty Changes)